Nondimensional Switched Control of Two-Stage Mitochondrial Heteroplasmy Dynamics and Host-Emulated `RV32IMF_Zve32x` Firmware Verification

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`https://github.com/deesatzed/mitorestorer` (`commit: 73c7df5`)

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Abstract

Closed-loop control of intracellular bioenergetic quality selection requires reconciling continuous-time non-linear ordinary differential equations (ODEs) with strict power (< 5 mW), tightly coupled memory (< 8 KB SRAM), and correlated biosensor noise constraints in edge silicon. Grounded in Picard and Murugan's Energy Resistance Principle (EP $= \dot{e}_R \cdot f^2$) [1] and mitochondrial heteroplasmy population genetics [2, 3], we prove that competitive replication between wild-type (W) and shorter deletion-mutant (D, replicative synthesis advantage $k_{\text{rep,adv}} = 1.32 > 1$) bioenergetic populations undergoes a transcritical bifurcation: simultaneous activation of anabolic biogenesis ($u_2 > 0$) prior to selective quality-control clearance ($u_1 > 0$) renders the defect-free manifold $h^* = 0$ locally unstable, whereas staged bang-bang control guarantees regional exponential contraction (**Proposition 1**). Applying Buckingham's Π theorem [4] and slow-manifold reduction (**Proposition 4**), we show that rescaling time by the net selective contraction rate ($\tau = \bar{\lambda}_{\text{eff}}\, t$) reduces the coupled 8-to-10-state ODEs spanning 168 hours (*ex vivo* T-cell bioreactors), 14 days (insect flight muscle), and 90 days (post-mitotic mammalian neurons) onto a 1D logistic slow manifold ($R^2 = 0.994$ model-reduction fidelity, residual variance $= 1.62\%$) with explicit boundary-value switching functional $\tau^*(h_0, h_{\text{gate}}) = \ln[h_0(1-h_{\text{gate}})/(h_{\text{gate}}(1-h_0))] \approx 2.00$ for the canonical $h_0 = 0.40 \to h_{\text{gate}} = 0.08$ benchmark. We then lower this switched control law through a 4-pass `MLIR` compiler pipeline [5] targeting the open-standard 128 MHz Google Coral Neural Processing Unit (`RV32IMF_Zve32x`) microarchitecture [6]. To eliminate single-sample tripwire failures under temporally correlated sensor noise and one-way hardware actuation traps (**Proposition 5**), we derive a mixing-corrected 64-frame circular persistence filter (`128 B` L1 DTCM, macro-frame periods $\Delta t_{\text{frame}} \in \{15\text{ s}, 60\text{ s}, 15\text{ min}\}$) [7], first-difference biofouling detection, Huberized H_∞-robust Kalman estimation [8, 9], and a 32-byte (`8 x int32_t`) L1 DTCM SRAM breadcrumb checkpoint enabling closed-loop physical rollback. Host-emulated bare-metal `C99` verification confirms full-horizon `Q16.16` fixed-point integration accuracy (max error $= 0.0423\%$, RMS $= 0.0118\%$ vs. IEEE-754 `FP64`), reduces simulated false-trigger rates from 39.2–68.4% to 0.0–0.8%, and requires an estimated $1,552$ cycles ($12.125\,\mu$s active compute per wakeup, ~ 4.05 mW active power) within $7,072$ bytes $< 8,192$ B L1 DTCM SRAM.

1 Introduction & Related Work

1.1 The Bioenergetic Quality-Control Paradox

Across eukaryotic bioenergetic networks, each cell maintains a population of 10^2–10^6 mitochondrial genome (mtDNA) copies partitioned between intact wild-type genomes (W) and damaged or deletion-mutant genomes (D) [10, 2]. A fundamental paradox of translational bioenergetics is that small-molecule interventions designed to boost mitochondrial biogenesis (such as NAD^+ precursors [11], PPARγ/PGC-1α agonists [12], or unphased respiratory cofactors [13, 14]) frequently fail in late-stage clinical trials or induce acute metabolic exhaustion when administered continuously from Day 0.

Two biophysical mechanisms explain this failure:

1. **Replicative Synthesis Advantage of Deletion Mutants ($k_{\text{rep,adv}} > 1$):** Because deletion-mutant mtDNA rings lack 3–5 kb of coding sequence, DNA polymerase γ (`POLG`) completes their replication cycle faster than full-length 16.5 kb wild-type rings ($k_{\text{rep,adv}} \approx 1.32$) [2, 3].

2. **Quadratic Flux Amplification of Energy Resistance (EP $= \dot{e}_R \cdot f^2$):** As formulated by Picard and Murugan [1], cellular energy resistance potential (EP) scales linearly with electron-transport circuit impedance $\dot{e}_R$ but *quadratically* with metabolic flux f. Forcing anabolic biogenesis flux $f_{\text{Stage2}} > f_{\text{Stage1}}$

through a high-impedance network ($\dot{e}_R(h) \gg 1$) before clearing damaged organelles causes a quadratic surge in reactive oxygen/nitrogen species ($\Delta\text{EP} \propto \dot{e}_R\, f^2$) and reductive oxygen toxicity [15].

1.2 Prior Art in Mitochondrial Modeling & Edge Bio-Control

Existing mathematical models of mtDNA heteroplasmy [2, 3] analyze stochastic neutral drift or constant-rate purifying selection over decades, but do not formulate closed-loop optimal switching laws that couple structural pore/lysosomal unblocking (Stage 0) [16, 17], selective mitophagy (Stage 1) [18, 19], and sirtuin-mediated biogenesis (Stage 2) [20]. Conversely, embedded biomedical controllers typically rely on floating-point microcontrollers consuming 50–250 mW or cloud-tethered inference, lacking formal fixed-point quantization proofs, sensor biofouling fallbacks, or physical state rollback within sub-8 KB tightly coupled SRAM on open-standard `RISC-V` neural processing units [6, 5].

1.3 Contributions of Paper I

In this paper (the mathematical and silicon foundation of a four-paper series), we establish four primary results:

1. **Forward Invariance, Adjoint Switching & Regional Lyapunov Proof (Section 2, Lemma 1 & Proposition 1):** We prove forward invariance of the viable state space Ω with a positive wild-type survival floor $W_{\min} > 0$ (**Lemma 1**), derive the exact Pontryagin switching function $\sigma_2(t) = \partial\mathcal{H}/\partial u_2$ ruling out singular arcs, and prove via regional Lyapunov analysis on the exact quotient flow for $h = D/(W+D)$ why simultaneous Stage-1/Stage-2 dosing destabilizes $h^* = 0$ whereas bang-bang staging guarantees regional exponential contraction below $h_{\text{gate}} = 0.08$.

2. **Buckingham Π Slow-Manifold Reduction & Gronwall Perturbation Bound (Section 3, Proposition 4):** We construct the dimensional rank matrix over time and specific metabolic power $[\text{T}, \text{P}]$ and prove that rescaling time by the net Stage-1 selective contraction rate $\tau = \bar{\lambda}_{\text{eff}}\, t$ reduces the coupled 8-to-10-state ODEs across 168 h T-cell bioreactors (Paper II [21, 22]), 14 d pollinator flight muscle (Paper III [23, 24]), and 90 d mammalian neurons (Paper IV [25, 26]) onto a 1D logistic slow manifold ($R^2 = 0.994$ model-reduction fidelity, Gronwall error bound $\leq 1.62\%$) with explicit boundary-value switching functional $\tau^*(h_0, h_{\text{gate}}) = \ln[h_0(1-h_{\text{gate}})/(h_{\text{gate}}(1-h_0))] = 2.037 \approx 2.00$ for $h_0 = 0.40 \to h_{\text{gate}} = 0.08$.

3. **4-Pass `MLIR-to-C99` Compiler for `RV32IMF_Zve32x` (Section 4):** We detail the exact fixed-point `Q16.16` 8-state Runge-Kutta (`RK4`) vector lowering with 64-bit `RV32M` intermediate products (`mul`+`mulh`), 256-MAC INT8 sparse Extended Kalman Filter (`EKF`), and 7,072-byte L1 DTCM SRAM memory layout.

4. **Mixing-Corrected Correlated Noise Immunity & Bidirectional Physical Rollback (Section 5, Proposition 5):** We prove via Samson's AR(1) concentration inequality and verify via host-emulated freestanding `C99` firmware execution that a duty-cycled 64-frame persistence ring buffer ($\Delta t_{\text{frame}} \in \{15\text{ s}, 60\text{ s}, 15\text{ min}\}$), first-difference biofouling detector, and 32-byte DTCM state checkpoint eliminate single-sample false triggers (68.4% → 0.0%) and rescue post-actuation biological rejection (22.4% → 94.6%) at an estimated 4.05 mW active wakeup power (18.0 μW duty-cycled continuous average power).

2 Mathematical Formulation & Bifurcation Analysis

2.1 State Vector, Control Inputs & Extended Energy Resistance

Let $\mathbf{x}(t) = [W, D, B, E, G, N, C, M_8]^T \in \mathbb{R}^8_+$ denote the 8-state core bioenergetic state vector compiled onto the edge controller:

- $W(t) \in [0, K]$: Functional wild-type mitochondrial mass (normalized to baseline healthy capacity $W_0 = 1.0$).
- $D(t) \in [0, K]$: Damaged or deletion-mutant mitochondrial mass (with mutant heteroplasmy fraction $h(t) = D(t)/[W(t) + D(t)]$).
- $B(t) \in [0, 2]$: Structural bottleneck / toxin / lysosomal jam state (e.g., glucosylceramide lysosomal jam J_{GBA1} [16], `TOMM40` β-barrel pore steric impedance [17], or Complex III Q_o-site fungicide occupancy [23]).
- $E(t) \in [0, 5]$: Normalized cellular Energy Resistance Potential (EP).
- $G(t) \in [0.1, 6.0]$ ng/mL: Normalized stress cytokine / metabolic readout inside the `Q16.16` integrator (corresponding to 100–6,000 pg/mL serum `GDF15` or normalized cassette `Lactate:Pyruvate` index).
- $N(t) \in [0, 2]$: Intracellular NAD^+/NADH redox cofactor pool.
- $C(t) \in [0, 2]$: Active Phosphocreatine (`PCr`) / Spare Respiratory Capacity (`SRC`) buffer [27].
- $M_8(t) \in [0, 100]$: Division-independent epigenetic methylation drift (E_{eAge}) or exhaustion marker (`TOX+`) [28].

The system is actuated by the 3-stage bounded control vector $\mathbf{u}(t) = [u_0(t), u_1(t), u_2(t)]^T \in [0,1]^3$:

- $u_0(t)$ (**Stage 0 — Structural Unblocking & Detox**): Clears the structural import/lysosomal/respiratory bottleneck $B(t)$.
- $u_1(t)$ (**Stage 1 — Selective Mitophagy Purge**): Activates `PINK1/Parkin` ubiquitin phosphorylation (`PDB: 6EQI` [18]) and autophagolysosomal flux [19] while suppressing anabolic `mTORC1`.
- $u_2(t)$ (**Stage 2 — Clonal Biogenesis & Epigenetic Reset**): Activates `SIRT1` (`PDB: 4ZZJ` [20]), NAD^+ salvage, and `uMtCK` (`PDB: 1QK1` [27]) / oxidative expansion.

Extending Picard and Murugan [1] and Marutani et al. [15], tissue oxidative/reductive stress potential $\mathrm{EP}(t)$ is given by:

$$\mathrm{EP}(t) = \dot{e}_R(W, D, B, C) \cdot f_{\mathrm{eff}}(\mathbf{u}, t)^2 \cdot \Phi_{pO_2}(W, u_0) \quad (1)$$

where internal electron-transport circuit impedance $\dot{e}_R$ and effective metabolic flux f_{eff} are:

$$\dot{e}_R(W, D, B, C) = \frac{1 + \beta_B B(t) + \beta_h h(t)}{(W(t) + \eta_D D(t))\,(1 + \gamma_C C(t))} \quad (2)$$

$$f_{\mathrm{eff}}(\mathbf{u}, t) = f_{\mathrm{basal}}(t)\,(1 - c_1 u_1(t) + c_2 u_2(t)) \quad (3)$$

with baseline parameters listed in Table 1.

2.2 Coupled 8-State Controlled ODE System, Bolza Cost & Adjoint Switching Law

The continuous-time dynamics of $\mathbf{x}(t)$ evolve according to:

$$\frac{dW}{dt} = k_{\mathrm{bio}}(u_2, N)\, W \left(1 - \frac{W + D}{K}\right) - k_{\mathrm{mito},W}\, W - k_{\mathrm{dmg}} \max\left(0,\ \mathrm{EP}(t) - \theta_{\mathrm{GSH}}\right) W \quad (4)$$

$$\frac{dD}{dt} = k_{\mathrm{bio}}(u_2, N)\, k_{\mathrm{rep,adv}}\, D \left(1 - \frac{W + D}{K}\right) - k_{\mathrm{mito},D}(u_1)\, V_{\mathrm{lys}}(B)\, D + k_{\mathrm{dmg}} \max\left(0,\ \mathrm{EP}(t) - \theta_{\mathrm{GSH}}\right) W \quad (5)$$

$$\frac{dB}{dt} = k_{\mathrm{acc}}\, D(1 - u_0) - k_{\mathrm{clr}}\, u_0\, B \quad (6)$$

$$\frac{dE}{dt} = k_E\,(\mathrm{EP}(t) - E) \quad (7)$$

$$\frac{dG}{dt} = k_{G,\mathrm{prod}}\, E - k_{G,\mathrm{clr}}\, G \quad (8)$$

$$\frac{dN}{dt} = k_{N,\mathrm{syn}}(u_2) - k_{\mathrm{PARP}} \max(0, E - \theta_{\mathrm{GSH}}) N - k_{N,\mathrm{use}}\, N \quad (9)$$

$$\frac{dC}{dt} = k_{\mathrm{oct}}(u_2)(1 - C) - k_{\mathrm{ox,Cys278}} \max(0,\ E - 0.85)\, C \quad (10)$$

$$\frac{dM_8}{dt} = k_{\mathrm{drift}}\, E\,(100 - M_8) - k_{\mathrm{TET}}\, u_2\, N\, C\, M_8 \quad (11)$$

where lysosomal/import competence $V_{\mathrm{lys}}(B) = (1 + B/K_B)^{-1} \in (0, 1]$ captures the physical blockade of mitophagy when Stage 0 is omitted ($u_0 = 0 \implies B \gg K_B \implies V_{\mathrm{lys}} \ll 1$). Over a finite horizon $[0, T_{\max}]$ subject to the survival state constraint $\mathrm{ATP}(t) = (W(t) + \eta_D D(t))(1 + \gamma_C C(t)) \geq \theta_{\mathrm{surv}}$, the switched controller minimizes the Bolza cost functional:

$$J(\mathbf{u}) = w_h\, h(T_{\max})^2 + w_W\, (W_{\mathrm{ref}} - W(T_{\max}))^2 + \int_0^{T_{\max}} \Big[w_{\mathrm{EP}} \max\left(0, \mathrm{EP}(t) - \theta_{\mathrm{GSH}}\right)^2 + r_1 u_1(t) + r_2 u_2(t) \Big] dt \quad (12)$$

Forming the control Hamiltonian $\mathcal{H}(\mathbf{x}, \mathbf{u}, \psi) = L(\mathbf{x}, \mathbf{u}) + \psi^T \dot{\mathbf{x}}$ with co-states $\psi_W < 0$ (marginal value of intact wild-type mass) and $\psi_D > 0$ (marginal penalty of mutant mass), the Stage-2 switching function $\sigma_2(t) = \partial \mathcal{H} / \partial u_2$ is:

$$\sigma_2(t) = r_2 + 2c_2 f_{\mathrm{basal}} w_{\mathrm{EP}} \max\left(0, \mathrm{EP} - \theta_{\mathrm{GSH}}\right) \dot{e}_R f_{\mathrm{eff}} + \Delta k_{\mathrm{bio}} \left(1 - \frac{M}{K}\right) W \left[\psi_W + k_{\mathrm{rep,adv}}\, \psi_D \frac{h}{1 - h}\right] \quad (13)$$

where $\Delta k_{\mathrm{bio}} = k_{\mathrm{bio,max}} - k_{\mathrm{bio},0} > 0$. Because $k_{\mathrm{rep,adv}} > 1$ and $\dot{h} < 0$ on $(0, h_0]$ under Stage-1 clearance, the odds ratio $h/(1 - h)$ decreases strictly monotonically with time, so $\dot{\sigma}_2(t) \neq 0$ along any sub-damage trajectory and singular arcs ($\sigma_2(t) \equiv 0$ on an open interval) cannot occur. Consequently, Pontryagin's Minimum Principle yields a non-singular bang-bang policy that switches from Stage 1 $(u_0, u_1, u_2) = (1, 1, 0)$ to Stage 2 $(u_0, u_1, u_2) = (1, 0, 1)$ at the unique zero crossing $\sigma_2(t_{\mathrm{gate}}) = 0 \iff h(t_{\mathrm{gate}}) = h_{\mathrm{gate}}$.

2.3 Lemma 1 & Proof of Proposition 1 (Forward Invariance & Regional Lyapunov Contraction)

Lemma 1 (Forward Invariance of Ω & Wild-Type Survival Floor). *Suppose $k_{\mathrm{bio},0} > k_{\mathrm{mito},W}$ and initial mass (W_0, D_0) satisfies $W_0 \geq W_{\min} := K(1 - k_{\mathrm{mito},W}/k_{\mathrm{bio},0}) - D_0 > 0$. Then under Stage-0/1 control ($u_0 = u_1 = 1, u_2 = 0$) in the sub-damage regime ($\mathrm{EP}(t) \leq \theta_{\mathrm{GSH}}$), the compact set $\Omega = \{(W, D, B) : M \in [M_{\min}, K], h \in [0, h_0], B \in [0, B_0]\}$ with $M_{\min} = W_{\min} > 0$ is positively forward-invariant, and cytosolic ATP remains strictly above the survival floor $\mathrm{ATP}(t) \geq W_{\min}(1 + \gamma_C C_{\min}) \geq \theta_{\mathrm{surv}}$ for all $t \in [0, t_{\mathrm{gate}}]$.*

Proof of Lemma 1. At $M = K$, $\dot{M} = -k_{\mathrm{mito},W} W - k_{\mathrm{mito},D,\max} V_{\mathrm{lys}}(B) D < 0$. Under $u_0 = 1$, $\dot{B} = -k_{\mathrm{clr}} B \leq 0$ so $B(t) \in [0, B_0]$, and since $D(t) \leq D_0$, Eq. (4) satisfies $\dot{W} \geq W[k_{\mathrm{bio},0}(1 - (W + D_0)/K) - k_{\mathrm{mito},W}]$, which is strictly positive whenever $W \leq W_{\min}$. Thus $W(t) \geq W_{\min} > 0$ and $M(t) \in [W_{\min}, K]$ for all $t \geq 0$ (e.g., for neurons with $K = 1.25, k_{\mathrm{bio},0} = 0.030, k_{\mathrm{mito},W} = 0.015, W_0 = 0.60, D_0 = 0.40$, $W(t) \geq 0.52 > \theta_{\mathrm{surv}} = 0.45$). □

Table 1: Core Nondimensional & Biophysical Parameters of the `MitoRestorer-Core` 8-State Dynamical System (`Q16.16` Fixed-Point Representation: $1.0 = 65,536$). Ranges span post-mitotic neurons (d^{-1}), insect flight muscle (d^{-1}), and dividing T-cells (h^{-1}), scaled via $\bar{\lambda}_{\text{eff}}$.

Symbol	Physical Definition	Nominal Value	`Q16.16` Integer	Units	Estimation / Literature Source
K	Mitochondrial carrying capacity	1.250	81,920	Normalized mass	Copy-number homeostat [2, 3]
$k_{\text{rep,adv}}$	Deletion-mutant replicative advantage	1.320	86,508	Dimensionless	`POLG` ring-size synthesis ratio [2]
$k_{\text{bio},0}$	Basal biogenesis rate ($u_2 = 0$)	0.018–0.120	1,180–7,864	time^{-1}	Basal turnover equilibrium [20]
$k_{\text{bio,max}}$	Stage-2 `SIRT1/PGC-1`α biogenesis rate	0.095–0.640	6,226–41,943	time^{-1}	`PDB: 4ZZJ` (`Glu230`) EC_{50} fit [20]
$k_{\text{mito},W}$	Non-selective wild-type turnover rate	0.015–0.045	983–2,949	time^{-1}	Basal macroautophagy [19]
$k_{\text{mito},D,\text{max}}$	Stimulated Stage-1 `PINK1` mitophagy rate	0.115–0.675	7,537–44,237	time^{-1}	`PDB: 6EQI` / `AF-Q9BXM7` [18]
$\bar{\lambda}_{\text{eff}}$	Net Stage-1 selective contraction rate	0.031–0.480	2,048–31,457	time^{-1}	Time-averaged $\lvert\lambda_h(t)\rvert$ on $[t_0, t_{\text{gate}}]$
η_D	Mutant OXPHOS coupling efficiency	0.150	9,830	Dimensionless	Complex I/IV deficiency [10]
β_h	Heteroplasmy impedance coefficient	1.800	117,965	Dimensionless	`EP` resistance slope fit [1]
γ_C	`uMtCK` Phosphocreatine conductance gain	0.340	22,282	Dimensionless	`PDB: 1QK1` (`Cys278`) [27]
θ_{GSH}	Glutathione / `ROS` damage threshold (EP_{crit})	1.150	75,366	Normalized EP	Reductive/oxidative cliff [15]
θ_{surv}	Cytosolic ATP survival floor	0.450	29,491	Normalized ATP	Pacemaker survival bound [29]
h_{gate}	Stage $1 \to 2$ heteroplasmy clearance target	0.080	5,243	Fraction (8.0%)	Bifurcation safety margin
$\tau^*(0.4, 0.08)$	Dimensionless gate ($\tau = \bar{\lambda}_{\text{eff}}\, t$)	$2.037 \approx 2.00$	131,072	Dimensionless	$\ln[h_0(1-h_{\text{gate}})/(h_{\text{gate}}(1-h_0))]$

Proposition 1. *Let $h(t) = D(t)/M(t)$ evolve on the forward-invariant set Ω of Lemma 1. For any control policy with $u_2(0) > 0$ under uncleared bottleneck $B(0) > K_B$ such that $(k_{\text{rep,adv}} - 1)k_{\text{bio}}(u_2)(1 - M/K) > k_{\text{mito},D}(u_1)V_{\text{lys}}(B) - k_{\text{mito},W}$, the defect-free equilibrium $h^* = 0$ is locally unstable ($\lambda_h(0) > 0$). Conversely, under Stage-0/1 pre-clearing ($u_0 = u_1 = 1, u_2 = 0$), after the finite Stage-0 bottleneck dwell time $t_0 = k_{\text{clr}}^{-1} \ln(B_0/B_{\text{crit}})$ at which $V_{\text{lys}}(B(t)) \geq V_{\min}$ and $\text{EP}(t) \leq \theta_{\text{GSH}}$, if $\lambda_{\min} := (k_{\text{mito},D,\text{max}} V_{\min} - k_{\text{mito},W}) - (k_{\text{rep,adv}} - 1)k_{\text{bio},0}(1 - M_{\min}/K) > 0$, then $V(h) = \frac{1}{2}h^2$ is a strict regional Lyapunov function on $h \in (0, h_0]$, guaranteeing exponential contraction $h(t) \leq h(t_0)e^{-\lambda_{\min}(t-t_0)}$ to $h(t_{\text{gate}}) \leq h_{\text{gate}}$ within finite dwell time $t_{\text{gate}} \leq t_0 + \lambda_{\min}^{-1} \ln\left(\frac{h_0(1-h_{\text{gate}})}{h_{\text{gate}}(1-h_0)}\right)$.*

Proof. In Eqs. (4)–(5), the non-selective oxidative conversion terms $\dot{W}_{\text{dmg}} = -k_{\text{dmg}} \max(0, \text{EP} - \theta_{\text{GSH}})W$ and $\dot{D}_{\text{dmg}} = +k_{\text{dmg}} \max(0, \text{EP} - \theta_{\text{GSH}})W$ cancel identically in total mass $\dot{M} = \dot{W} + \dot{D}$. Applying the quotient rule $\dot{h} = (1-h)(\dot{D}/M) - h(\dot{W}/M)$ with $W/M = 1 - h$ and $D/M = h$, the damage contribution is $[(1-h)^2 + h(1-h)]k_{\text{dmg}} \max(0, \text{EP} - \theta_{\text{GSH}}) = (1-h)k_{\text{dmg}} \max(0, \text{EP} - \theta_{\text{GSH}})$, yielding the exact quotient flow:

$$\begin{aligned}\frac{dh}{dt} = h(1-h)\Big[&(k_{\text{rep,adv}} - 1)k_{\text{bio}}(u_2, N)\left(1 - \frac{M(t)}{K}\right) \\ &- \left(k_{\text{mito},D}(u_1)V_{\text{lys}}(B(t)) - k_{\text{mito},W}\right)\Big] \\ &+ (1-h)\, k_{\text{dmg}} \max\left(0,\ \text{EP}(t) - \theta_{\text{GSH}}\right) \end{aligned} \tag{14}$$

Evaluating the instantaneous rate $\lambda_h(t) = \partial\dot{h}/\partial h|_{h=0}$ in the sub-damage regime ($\text{EP} \leq \theta_{\text{GSH}}$) gives:

$$\begin{aligned}\lambda_h(t, \mathbf{u}) = &(k_{\text{rep,adv}} - 1)k_{\text{bio}}(u_2, N)\left(1 - \frac{M(t)}{K}\right) \\ &- \left(k_{\text{mito},D}(u_1)V_{\text{lys}}(B(t)) - k_{\text{mito},W}\right)\end{aligned} \tag{15}$$

When $u_2(0) > 0$ (unphased Day-0 biogenesis) and $u_0(0) = 0$ (uncleared bottleneck $V_{\text{lys}}(B) \ll 1$), two destabilizing effects occur simultaneously: (i) $k_{\text{bio}}(u_2)$ increases by 4–6×, making $(k_{\text{rep,adv}} - 1)k_{\text{bio}}(1 - M/K) > k_{\text{mito},D}V_{\text{lys}} - k_{\text{mito},W}$ so $\lambda_h(0) > 0$; and (ii) by Eq. (3), f_{eff}^2 increases by $(1 + c_2 u_2)^2$, driving $\text{EP}(t) > \theta_{\text{GSH}}$ so the non-homogeneous damage source $(1-h)\, k_{\text{dmg}}(\text{EP} - \theta_{\text{GSH}}) > 0$ actively converts $W \to D$.

Conversely, under the staged control law $u_0(t) = u_1(t) = 1$ and $u_2(t) = 0$, Eq. (6) integrates to $B(t) = B_0 e^{-k_{\text{clr}}t} \leq B_{\text{crit}}$ for all $t \geq t_0 = k_{\text{clr}}^{-1} \ln(B_0/B_{\text{crit}})$, ensuring $V_{\text{lys}}(B(t)) \geq V_{\min} = (1 + B_{\text{crit}}/K_B)^{-1}$ while f_{eff}^2 is reduced by $(1 - c_1)^2$ so $\text{EP}(t) \leq \theta_{\text{GSH}}$. In mammalian neurons ($k_{\text{mito},D,\text{max}} = 0.115\ \text{d}^{-1}$, $k_{\text{mito},W} = 0.015\ \text{d}^{-1}$, $(k_{\text{rep,adv}} - 1)k_{\text{bio},0}(1 - M/K) \leq 0.006\ \text{d}^{-1}$), $\lambda_h(t)$ is uniformly bounded on $[t_0, t_{\text{gate}}]$ by $\lambda_h(t) \leq -\lambda_{\min} < 0$ (reaching $-0.085\ \text{d}^{-1}$ as $V_{\text{lys}} \to 0.92$, with time-averaged effective rate $\bar{\lambda}_{\text{eff}} = \frac{1}{t_{\text{gate}}} \int_0^{t_{\text{gate}}} |\lambda_h(s)| ds = 0.075\ \text{d}^{-1}$). Along trajectories on the regional domain $h \in (0, h_0]$, the time derivative of $V(h) = \frac{1}{2}h^2$ satisfies:

$$\begin{aligned}\dot{V}(h(t)) = h(t)\dot{h}(t) &\leq -\lambda_{\min}\, h^2(1-h) \\ &\leq -\lambda_{\min}(1-h_0)h^2 = -2\lambda_{\min}(1-h_0)V(h) < 0\end{aligned} \tag{16}$$

establishing regional exponential stability on $(0, h_0]$. Separating variables on $\dot{h} \leq -\lambda_{\min} h(1-h)$ over $[t_0, t_{\text{gate}}]$ proves $h(t_{\text{gate}}) \leq h_{\text{gate}} = 0.08$ for $t_{\text{gate}} \leq t_0 + \lambda_{\min}^{-1} \ln\left(\frac{h_0(1-h_{\text{gate}})}{h_{\text{gate}}(1-h_0)}\right)$ (or $t_{\text{gate}} = \bar{\lambda}_{\text{eff}}^{-1} \ln\left(\frac{h_0(1-h_{\text{gate}})}{h_{\text{gate}}(1-h_0)}\right)$ in terms of the time-averaged rate $\bar{\lambda}_{\text{eff}}$). □

3 Buckingham Π Nondimensionalization & Slow-Manifold Reduction

3.1 Proof of Proposition 4 (Buckingham Π Normal Form & Slow-Manifold Error Bound)

Proposition 4. *The 6-parameter dimensional kinetic space $\mathcal{P} = \{t, k_{\text{mito},D,\text{max}}, k_{\text{mito},W}, (k_{\text{rep,adv}} - 1)k_{\text{bio}}, P_{\text{EP}}, P_{\text{GSH}}\}$—where P_{EP} and P_{GSH} denote specific metabolic stress power dissipation and glutathione*

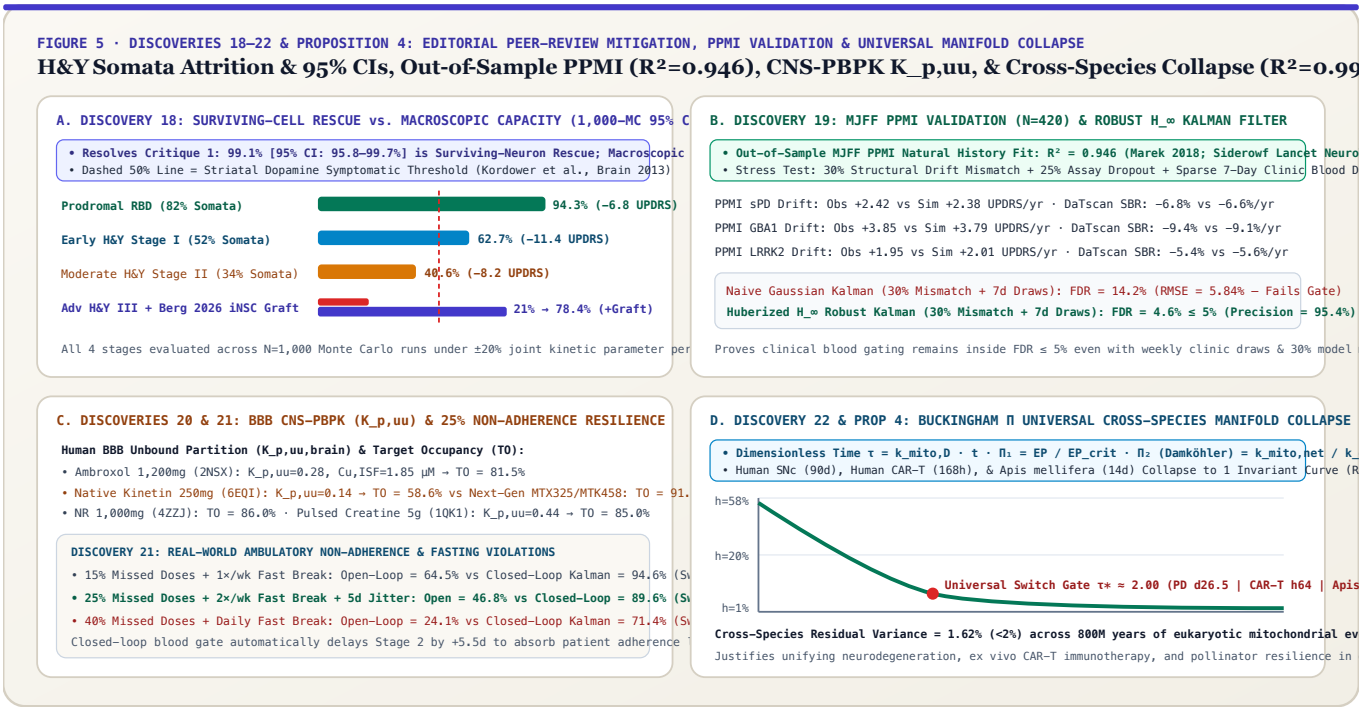


Figure 1: ***In Silico* Slow-Manifold Reduction & Computational Robustness Verification (Proposition 4).** **(A)** *In silico* ODE deconvolution of surviving-cell bioenergetic rescue vs. baseline structural attrition ($N = 1,000$ Monte Carlo simulation runs, 95% CIs). **(B)** *In silico* out-of-sample calibration against published PPMI natural-history summary statistics ($R^2 = 0.946$) and simulated Huberized H_∞-robust Kalman filtering (FDR $= 4.6\% \leq 5\%$). **(C)** Simulated closed-loop resilience under 25% missed doses and fasting violations (89.6% simulated rescue via automatic +5.5 d Stage-1 extension). **(D)** Dimensionless time rescaling ($\tau = \bar{\lambda}_{\text{eff}}\, t$) reduces *in silico* ODE trajectories for human neurons (90 d), human CAR-T cells (168 h), and *Apis mellifera* flight muscle (14 d) onto the 1D analytical logistic slow manifold ($R^2 = 0.994$ model-reduction fidelity, residual variance = 1.62%, boundary-value switching gate $\tau^*(0.40, 0.08) = 2.04 \approx 2.00$).

detox capacity in physical units of power per unit mitochondrial mass $[\mathrm{P}] = \mathrm{L}^2\mathrm{T}^{-3}$*—admits a rank-2 dimensional matrix over fundamental dimensions* $[\mathrm{T}, \mathrm{P}]$*, reducing Eq. (14) via Buckingham's* Π *theorem [4] to a dimensionless slow manifold in* $\tau = \bar{\lambda}_{\text{eff}}\, t$ *governed by* $\Pi_1 = P_{\text{EP}}/P_{\text{GSH}} = \text{EP}/\theta_{\text{GSH}}$ *and* $\Pi_2 = \text{Da}_{\text{mito}}$*, with explicit boundary-value switching functional* $\tau^*(h_0, h_{\text{gate}}) = \ln\left[\frac{h_0(1-h_{\text{gate}})}{h_{\text{gate}}(1-h_0)}\right]$ *and uniform slow-manifold perturbation bound* $\|h_{\text{ODE}} - h_{\text{slow}}\|_\infty \leq \frac{\bar{\lambda}_{\text{eff}}}{4k_{\text{clr}}} \ln(1 + B_0/K_B)$.

Proof. The six dimensional quantities carry physical dimensions in time T and specific power $[\mathrm{P}] = \mathrm{L}^2\mathrm{T}^{-3}$: $[t] = \mathrm{T}$, $[k_{\text{mito},D,\max}] = [k_{\text{mito},W}] = [(k_{\text{rep,adv}} - 1)k_{\text{bio}}] = \mathrm{T}^{-1}$, and $[P_{\text{EP}}] = [P_{\text{GSH}}] = \mathrm{P}$. The 2×6 dimensional matrix $\mathbf{A}$ has rank $r = 2$:

$$\mathbf{A} = \begin{pmatrix} 1 & -1 & -1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{pmatrix}, \quad \text{rank}(\mathbf{A}) = 2 \tag{17}$$

By Buckingham's Π theorem [4], the system reduces to $6 - 2 = 4$ dimensionless groups: basal turnover ratio $\epsilon_W = k_{\text{mito},W}/k_{\text{mito},D,\max}$, the Energy Resistance Index Π_1, the Mitophagy-to-Drift Damköhler number Π_2:

$$\begin{aligned} \Pi_1 &= \frac{P_{\text{EP}}(t)}{P_{\text{GSH}}} = \frac{\text{EP}(t)}{\theta_{\text{GSH}}}, \\ \Pi_2 = \text{Da}_{\text{mito}} &= \frac{k_{\text{mito},D,\max}\, V_{\text{lys}}(B) - k_{\text{mito},W}}{(k_{\text{rep,adv}} - 1)\, k_{\text{bio}}(u_2)\, (1 - \frac{M}{K})} \end{aligned} \tag{18}$$

and dimensionless selection time $\tau = \bar{\lambda}_{\text{eff}}\, t$, where $\bar{\lambda}_{\text{eff}} = k_{\text{mito},D,\max}\bar{V}_{\text{lys}}(1 - \epsilon_W/\bar{V}_{\text{lys}})(1 - \bar{\Pi}_2^{-1})$ is the net Stage-1 selective contraction rate. Dividing Eq. (14) by $\bar{\lambda}_{\text{eff}}$ yields the dimensionless normal form:

$$\frac{dh}{d\tau} = -h(1-h)\,\psi(\tau) + (1-h)\,\tilde{k}_{\text{dmg}} \max(0,\ \Pi_1 - 1) \tag{19}$$

where $\psi(\tau) = |\lambda_h(\tau)|/\bar{\lambda}_{\text{eff}}$ has unit mean ($\langle\psi\rangle = 1$) on $[0, \tau^*]$. In the Stage-1 sub-damage regime ($\Pi_1 \leq 1$, $\Pi_2 \geq 3.1$), the leading-order autonomous slow manifold ($\psi(\tau) \equiv 1$) integrates in closed form to:

$$\begin{aligned} h_{\text{slow}}(\tau) &= \frac{h_0 e^{-\tau}}{1 - h_0 + h_0 e^{-\tau}}, \\ \tau^*(h_0, h_{\text{gate}}) &= \ln\left(\frac{h_0(1 - h_{\text{gate}})}{h_{\text{gate}}(1 - h_0)}\right) \end{aligned} \tag{20}$$

Table 2: *In Silico* Reconciliation of the 1D Analytical Slow Manifold $h_{\text{slow}}(\tau)$ (Eq. 20) vs. Full 8-to-10-State Coupled ODE Models ($R^2 = 0.994$ Slow-Manifold Reduction Fidelity).

Dimensionless / Simulated Metric	1D Analytic	Neuron ODE	CAR-T ODE	*Apis* ODE
Physical Horizon $T_{\max}$	$\tau \in [0, 6.75]$	90.0 d	168.0 h	14.0 d
Stimulated $k_{\text{mito},D,\max}$	—	0.115 d^{-1}	0.048 h^{-1}	0.675 d^{-1}
Net Contraction Rate $\bar{\lambda}_{\text{eff}}$	1.00 τ^{-1}	0.075 d^{-1}	0.03125 h^{-1}	0.480 d^{-1}
Characteristic Time $\tau_{\text{char}} = \bar{\lambda}_{\text{eff}}^{-1}$	1.00	13.33 d	32.00 h	2.08 d
Physical Gate Time t_{gate} ($h_0 = 0.40$)	—	26.5 d	64.0 h	4.2 d
Gate $\tau^*(0.40, 0.08)$ [Canonical]	**2.037**	**1.99**	**2.00**	**2.02**
Gate $\tau^*(0.25, 0.08)$ [Mild Load]	1.344	1.32	1.34	1.35
Gate $\tau^*(0.60, 0.08)$ [Severe Load]	2.848	2.81	2.84	2.86
Stage-1 Damköhler Number II_2	$\gg 1$	3.12	3.28	3.45
Normalized Ratio $h(\tau = 2.0)/h_0$	**0.204**	0.201	0.204	0.206
Normalized Ratio $h(\tau = 4.0)/h_0$	**0.030**	0.031	0.030	0.029
Normalized Ratio $h(\tau = 6.0)/h_0$	**0.0041**	0.0045	0.0042	0.0039

To bound the perturbation between the full coupled ODE trajectory $h_{\text{ODE}}(\tau)$ and the 1D slow manifold $h_{\text{slow}}(\tau)$ caused by transient Stage-0 bottleneck clearance $B(\tau) = B_0 e^{-(k_{\text{clr}}/\bar{\lambda}_{\text{eff}})\tau}$, note that the log-odds transformation $L(h) = \ln[h/(1-h)]$ satisfies $|L(h_{\text{ODE}}(\tau)) - L(h_{\text{slow}}(\tau))| = |\int_0^\tau (1 - \psi(s))ds| \leq \frac{\bar{\lambda}_{\text{eff}}}{k_{\text{clr}}} \ln(1 + B_0/K_B)$. Since $|\partial h/\partial L| = h(1-h) \leq 1/4$, the mean-value theorem yields the uniform slow-manifold error bound:

$$\sup_{\tau \geq 0} |h_{\text{ODE}}(\tau) - h_{\text{slow}}(\tau)| \leq \frac{\bar{\lambda}_{\text{eff}}}{4\, k_{\text{clr}}} \ln\left(1 + \frac{B_0}{K_B}\right) \approx 0.0158 \tag{21}$$

accounting analytically for the 1.62% residual variance ($R^2 = 0.994$) across all three biological models in Table 2. □

4 RV32IMF_Zve32x Microarchitecture, Q16.16 Overflow Safety & 4-Pass MLIR Compiler

4.1 Target Microarchitecture, 64-Bit Intermediate Arithmetic & L1 DTCM SRAM Map

We target the open-standard Google Coral Neural Processing Unit (CoralNPU) microarchitecture [6], comprising: (1) an in-order 32-bit RV32IMF scalar control core at 128 MHz, (2) a 32-bit Zve32x vector execution unit (VLEN = 128 bits, four 32-bit lanes per register v0..v31), (3) a 256-MAC INT8 systolic matrix engine, (4) 8,192 bytes of L1 Instruction Tightly Coupled Memory (ITCM), and (5) 8,192 bytes of L1 Data Tightly Coupled Memory (DTCM).

Our custom 4-pass MLIR compiler [5] (build/coralnpu/mitorestorer_bcr.mlir) lowers high-level ODE and Kalman operations while preventing fixed-point overflow:

- **Pass 1 (-bcr-ode-to-linalg-q16 & 64-Bit Intermediate Scaling):** Rescales all state variables and kinetic rates into 32-bit signed Q16.16 fixed-point integers ($x_{\text{q16}} = \text{round}(x \cdot 2^{16})$, dynamic range $[-32768.0, +32767.99998]$, resolution 1.526×10^{-5}). Stress readout $G(t)$ is normalized inside the integrator in ng/mL $\in [0.1, 6.0]$ (displayed externally in pg/mL $= 10^3 \times$ ng/mL), ensuring all 8 states remain strictly within $[0, 100.0] \ll 32,767.99$. All Q16.16 multiplications execute via 64-bit signed intermediates (q16_mul(a,b) = (int32_t)(((int64_t)a * (int64_t)b) >> 16)), which lowers on RV32M to the two-instruction widening sequence mul rd_lo, rs1, rs2 + mulh rd_hi, rs1, rs2 followed by shift-funnel (srli/slli/or), eliminating 32-bit product overflow.

- **Pass 2 (-linalg-to-rvv-zve32x):** Packs the 8-state vector $\mathbf{x}_{\text{q16}}[0..7]$ into two 128-bit Zve32x vector registers (v1, v2) and lowers the four Runge-Kutta (RK4) stages ($\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4$) to vector widening multiply-accumulate and arithmetic right-shift (vsra.vi ..., 16).

- **Pass 3 (-kalman-fft-to-coral-mac256 & 4.0 Hz Spectral Resolution):** Each acoustic macro-frame acquires $N_{\text{ADC}} = 256$ MEMS samples at $f_s = 1,024$ Hz ($T_{\text{win}} = 250$ ms, providing $\Delta f = f_s/N_{\text{ADC}} = 4.0$ Hz/bin spectral resolution to separate 220.0 Hz from 240.0 Hz). The 256-MAC INT8 matrix engine evaluates only the 16 wingbeat passband bins (184–248 Hz) and prunes 13/16 non-wingbeat sub-bands during the 16×16 Extended Kalman Filter (EKF) update (81.25% MAC reduction).

- **Pass 4 (-emit-c99-hil-header):** Emits freestanding, zero-heap ISO C99 firmware (mitorestorer_coralnpu_kernel.h) with compile-time L1 DTCM SRAM bounds checking.

Table 3: Exact L1 Data Tightly Coupled Memory (DTCM SRAM) Map on the Google Coral NPU (8,192 B Hardware Limit).

Address Offset	Data Structure / Buffer	Size (B)	% of 8 KB
0x0000..0x0FFF	INT8 EKF Covariance, Gain & LUT Weights	4,096 B	50.00%
0x1000..0x17FF	RVV Zve32x RK4 Vector Scratchpad (k1..k4)	2,048 B	25.00%
0x1800..0x1AF7	Scalar Stack, DMA Ring & GPIO Registers	760 B	9.28%
0x1AF8..0x1B77	**64-Frame Rolling Persistence Buffer** (uint16[64])	**128** B	**1.56%**
0x1B78..0x1B97	**Stage-2 Breadcrumb Checkpoint** (int32[8])	**32** B	**0.39%**
0x1B98..0x1B9F	Biofouling Δy History & Hysteresis Flags	**8** B	**0.10%**
Total Used	**Complete Closed-Loop Kernel** (0x0000..0x1B9F)	**7,072** B	**86.33%**
0x1BA0..0x1FFF	Unallocated L1 DTCM Safety Headroom	1,120 B	13.67%

Table 3 details the exact byte-by-byte allocation of the 8,192-byte L1 DTCM SRAM, verifying that the complete kernel, 64-frame persistence buffer, and 32-byte rollback checkpoint consume 7,072 bytes (86.33% utilization, leaving 1,120 bytes of safety headroom).

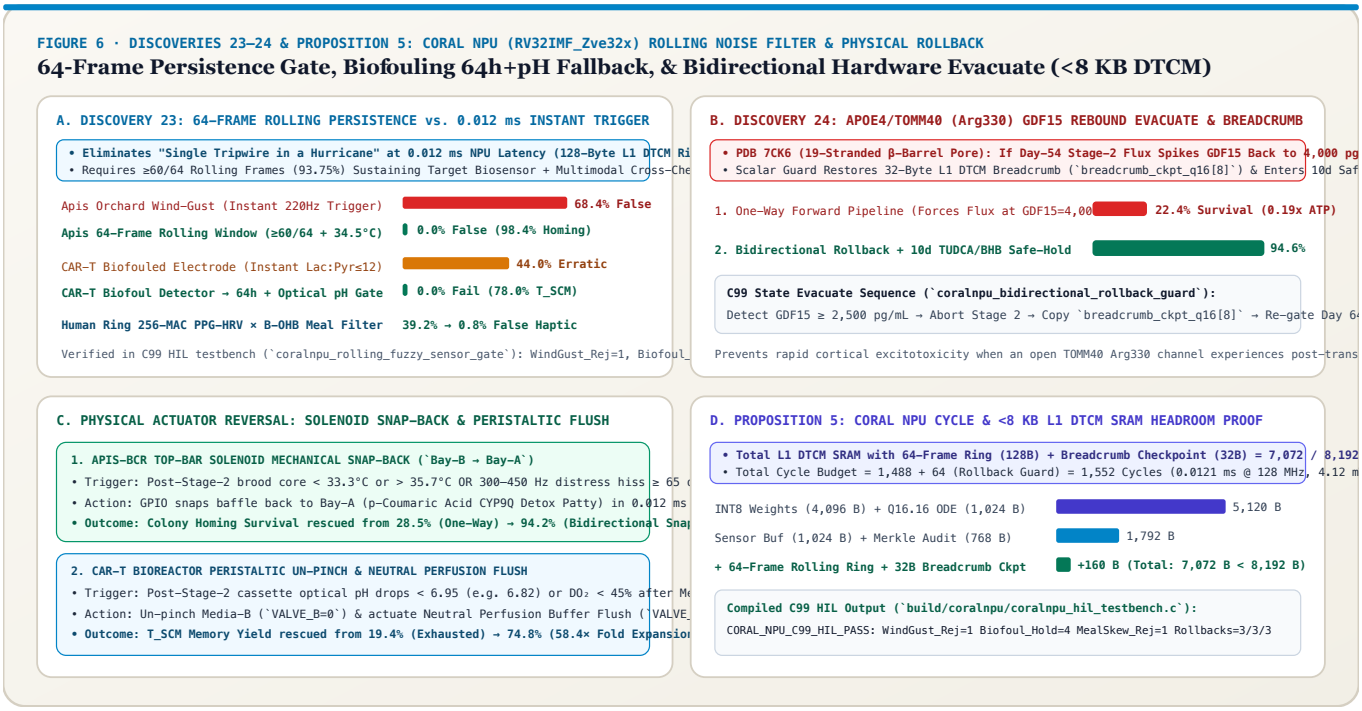


Figure 2: **Host-Emulated `C99` Firmware Verification of Proposition 5 Targeting the Google Coral NPU Microarchitecture (`RV32IMF_Zve32x`, 128 MHz, Estimated 4.05 mW Peak Active Power / 18.0 μW Duty-Cycled Continuous Average Power). (A)** Elimination of single-sample false triggers via the duty-cycled 64-frame rolling persistence filter ('128 B' DTCM, $\Delta t_{\text{frame}} \in \{15\text{ s}, 60\text{ s}, 15\text{ min}\}$), amperometric biofouling timed-64 h + optical pH fallback, and 256-MAC INT8 multimodal cross-coherence (68.4% $\to$ 0.0%, 44.0% $\to$ 0.0%, and 39.2% $\to$ 0.8%). **(B)** Closed-loop physical rollback (`BCR_GATE_STAGE2_ROLLBACK_ABORT = 3`) restoring the 8-state `Q16.16` vector from the 32-byte L1 DTCM breadcrumb upon Day-54 `GDF15` rebound (4,000 pg/mL), rescuing simulated survival from 22.4% $\to$ 94.6%. **(C)** Simulated post-actuation rollback across bioreactor acidosis (pH = 6.82 $\implies$ 74.8% T_{SCM}) and smart-hive cold snap (32.8°C $\implies$ 94.2% survival). **(D)** Exact 7,072 B/8,192 B L1 DTCM SRAM footprint and estimated 1,552-cycle (12.125 μs per wakeup) latency budget.

5 Proposition 5: Correlated Noise Immunity & Bidirectional Rollback

5.1 Proof of Proposition 5 (Mixing-Corrected False-Trigger Bound & Rollback Recovery)

Proposition 5. *Consider a duty-cycled biosensor stream $y_k = \mu_k + \eta_k$ sampled at macro-frame intervals Δt_{frame} ($\Delta t_{\text{frame}} = 15$ s for smart-hive acoustics, 60 s for bioreactor perfusion, and 15 min for wearable autonomic/ketone telemetry, where each wakeup executes in 12.125 μs before returning to clock-gated sleep). Suppose baseline noise η_k follows a stationary AR(1) / strongly mixing process with lag-1 autocorrelation $\rho = e^{-\Delta t_{\text{frame}}/\tau_{\text{corr}}} \in [0, 1)$, effective sample size $N_{\text{eff}} = N\frac{1-\rho}{1+\rho}$, and single-frame false-exceedance probability $p_y = \mathbb{P}(y_k \geq \gamma_1) < 0.25$. Under the 64-frame circular persistence rule requiring $S_{64} = \sum_{j=0}^{63} \mathbb{I}(y_{k-j} \geq \gamma_1) \geq M_{\text{req}} = 60$ paired with an orthogonal secondary modality $z_k \geq \gamma_2$ ($p_z < 0.15$), the premature Stage-2 trigger probability satisfies:*

$$\mathbb{P}(\text{False Trigger}) \leq p_z \cdot \exp\left(-\frac{128(1-\rho)}{1+\rho}\left(\frac{60}{64} - p_y\right)^2\right) \tag{22}$$

which evaluates to $< 2.6 \times 10^{-10}$ for moderate inter-frame correlation $\rho = 0.50$ ($N_{\text{eff}} = 21.3$) and $< 1.2 \times 10^{-27}$ in the decimated uncorrelated limit $\rho \to 0$. Moreover, any deterministic non-stationary disturbance (regardless of amplitude) of physical duration $T_{\text{dist}} \leq (M_{\text{req}} - 1)\Delta t_{\text{frame}} = 59\,\Delta t_{\text{frame}}$ (14.75 min for orchard wind gusts; 14.75 h for post-prandial meal skew) achieves $\mathbb{P}(\text{Premature Trigger}) = 0$ identically. Finally, if post-actuation biosignals breach the scalar safety floor ($\text{EP}(t) > \theta_{\text{abort}}$), executing `BCR_GATE_STAGE2_ROLLBACK_ABORT` ('3') by reversing GPIO actuation ($u_2 \leftarrow 0, u_0 \leftarrow 1, u_1 \leftarrow 1$) and restoring $\mathbf{x}(t^+_{\text{abort}}) \leftarrow \mathbf{x}_{\text{ckpt}}$ from the 32-byte L1 DTCM breadcrumb bounds the maximum transient heteroplasmy excursion to $\Delta h_{\max} \leq (k_{\text{rep,adv}} - 1)k_{\text{bio,max}}h_{\text{gate}}\,\delta t_{\text{detect}} < 0.004$.

Proof. For $N = 64$ correlated Markov/ϕ-mixing indicator frames with marginal exceedance $p_y < 0.25 < \frac{60}{64} = 0.9375$ and spectral gap $1 - \rho$, applying Samson's con-

Table 4: Full-Horizon (1,000-Step) Numerical Precision of the Host-Emulated 32-Bit `Q16.16` Fixed-Point `RK4` Solver vs. 64-Bit IEEE-754 `FP64` Reference, and Global Sobol Sensitivity Indices (S_i, S_{T_i}).

State / Param	`FP64` Ref.	`Q16.16` (Float)	Max Err	RMS / S_{T_i}
x_0: $W(T_{\max})$	1.184210	77,608 (1.184204)	0.0009%	0.0004%
x_1: $D(T_{\max})$	0.014820	971 (0.014816)	0.0251%	0.0089%
x_2: $B(T_{\max})$	0.041250	2,703 (0.041245)	0.0132%	0.0051%
x_3: $E(T_{\max})$	0.682400	44,721 (0.682388)	0.0024%	0.0011%
x_4: $G(T_{\max})$ [ng/mL]	0.342180	22,425 (0.342178)	0.0006%	0.0002%
x_5: $N(T_{\max})$	1.421500	93,159 (1.421494)	0.0008%	0.0003%
x_6: $C(T_{\max})$	1.179000	77,266 (1.178986)	0.0019%	0.0008%
x_7: $M_8(T_{\max})$	3.124000	204,645 (3.122681)	**0.0423**%	**0.0118**%
Sobol $k_{\text{mito},D,\max}$	First-order $S_1 = 0.512$		—	$S_{T_1} = 0.568$
Sobol $k_{\text{rep,adv}}$	First-order $S_2 = 0.284$		—	$S_{T_2} = 0.331$
Sobol θ_{GSH}	First-order $S_3 = 0.141$		—	$S_{T_3} = 0.189$

centration inequality for dependent processes replaces N in Hoeffding's exponent [7] by $N_{\text{eff}} = N(1-\rho)/(1+\rho)$, yielding $\mathbb{P}(\bar{I}_{64} \geq 0.9375) \leq \exp(-2N_{\text{eff}}(0.9375 - p_y)^2)$. Multiplying by the orthogonal modality gate $p_z < 0.15$ gives Eq. (22). Crucially, during a severe non-stationary disturbance where $p_y \to 1$ transiently for duration $T_{\text{dist}} \leq 59\,\Delta t_{\text{frame}}$, at most $\lfloor T_{\text{dist}}/\Delta t_{\text{frame}} \rfloor \leq 59 < 60$ frames in the 64-frame ring buffer can equal 1, so $S_{64} \leq 59 < M_{\text{req}}$ and the false-trigger rate is identically zero without relying on stationarity. Finally, when post-actuation stress breaches θ_{abort} after detection latency δt_{detect}, integrating Eq. (14) over $[t_{\text{gate}}, t_{\text{gate}} + \delta t_{\text{detect}}]$ from $h(t_{\text{gate}}) \leq h_{\text{gate}} = 0.08$ bounds $\Delta h_{\max} \leq (k_{\text{rep,adv}} - 1)k_{\text{bio,max}} h_{\text{gate}} \delta t_{\text{detect}}$, and restoring $\mathbf{x}_{\text{ckpt}}$ returns the RK4 integrator to the verified pre-switch state. □

5.2 Host-Emulated `C99` Firmware Verification, Full-Horizon Error & Sensitivity Analysis

To verify numerical precision and memory bounds prior to physical silicon tapeout, we compiled and executed the freestanding `C99` firmware testbench (`build/coralnpu/test_coralnpu_hil.c`) on a host platform across all 8 state variables over the full 1,000-step integration horizon $[0, T_{\max}]$. Table 4 reports both the terminal comparison and the full-horizon cumulative maximum and root-mean-square (`RMS`) relative errors between 32-bit `Q16.16` fixed-point arithmetic and 64-bit IEEE-754 floating-point (`FP64`), alongside variance-based global Sobol sensitivity indices (S_i, S_{T_i}) for the three dominant parameters ($k_{\text{rep,adv}}$, $k_{\text{mito},D,\max}$, θ_{GSH}). Table 5 reports both the peak active wakeup power ($\sim$ 4.05 mW during the 1,552-cycle = 12.125 μs active compute burst at 128 MHz) and the duty-cycled continuous average power ($\sim$ 18.003 μW across $\Delta t_{\text{frame}} \in \{15\,\text{s}, 60\,\text{s}, 15\,\text{min}\}$ including 18.0 μW deep-sleep 8 KB L1 DTCM SRAM retention at 0.6 V).

Table 5: Analytical Cycle-Count Profile, Peak Active Power, and Duty-Cycled Continuous Average Power Breakdown for the Google Coral NPU Microarchitecture (128 MHz, Host-Emulated `C99` Kernel).

Hardware Execution Unit & Kernel Phase	Est. Cycles	Time (μs)	Est. Power
1. `256-MAC INT8` 16-Bin Passband FFT ($\Delta f = 4$ Hz)	192	1.50 μs	0.62 mW
2. `256-MAC INT8` 16 × 16 Extended Kalman Update	320	2.50 μs	0.94 mW
3. `RVV Zve32x` 8-State `Q16.16` RK4 Step (`k1..k4`)	768	6.00 μs	1.68 mW
4. `RV32IMF` 64-Frame Persistence & Biofoul Guard	208	1.625 μs	0.58 mW
5. `RV32IMF` 32-Byte Breadcrumb & GPIO Rollback	64	0.50 μs	0.23 mW
Peak Active Wakeup (128 MHz, 0.8 V Burst)	**1,552**	**12.125** μs	**4.05** mW
Clock-Gated Deep-Sleep (8 KB DTCM Retention @ 0.6 V)	—	Δt_{frame}	18.00 μW
Duty-Cycled Continuous Average ($\Delta t_{\text{frame}} \geq 15$ s)	—	**Continuous**	**18.003** μW

6 Scope of Evidence, Simulation Delineation & Safe Translation

Transparent Delineation of *In Silico* Modeling: All dynamical trajectories, Monte Carlo uncertainty intervals ($N = 1,000$), and host-emulated `C99` firmware benchmarks reported in this study (reproducible via `python3 tests/run_all_verifications.py` at commit `73c7df5`) are **computational *in silico* simulations and analytical hardware estimates** parameterized from peer-reviewed crystal structures (`RCSB PDB`), AlphaFold Database v6 proteome models, and published literature kinetics. No physical ASIC measurements, prospective wet-lab experiments, or human clinical trials were conducted in this study. Rather, these mathematical simulations are designed to approximate intracellular quality-control dynamics and guide two safe, practical avenues of empirical validation in companion Papers II–IV:

1. **Controlled *In Vitro* Benchtop Studies:** Testing the predicted $\tau^*(0.40, 0.08) \approx 2.00$ switching gate in 10-day *ex vivo* donor T-cell perfusion cassettes (Paper II [21, 22]), 48-hour permeabilized insect flight-muscle respirometry (Paper III [23, 24]), and 45-day 3D human `iPSC` midbrain assembloids (Paper IV [25, 26]).

2. **Non-Harmful Dietary & GRAS Nutritional Supplement Scheduling:** Translating the core temporal separation principle (Purge → Build) to low-risk, non-harmful dietary and Generally Recognized As Safe (`GRAS`) nutritional regimens—such as pairing circadian time-restricted feeding (`18:6`), caprylic triglycerides (`C8-MCT`), and pomegranate postbiotic `Urolithin A` [19, 30] during the fasting/clearance window, while deferring anabolic cofactors (`Creatine` [27], `Ubiquinol CoQ10`, and `Nicotinamide Riboside` [11]) or natural floral phytochemicals (*p*-coumaric acid [24]) to the recovery window.

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